

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/303531566>

Maximal aerobic power and anaerobic capacity in cycling across the age spectrum in male master athletes

Article in *European Journal of Applied Physiology* · May 2016

DOI: 10.1007/s00421-016-3396-9

CITATIONS

7

READS

621

5 authors, including:



Carlo Capelli

Norwegian School of Sport Sciences (NIH)

142 PUBLICATIONS 2,567 CITATIONS

[SEE PROFILE](#)



Paolo Bruseghini

Università degli Studi di Brescia

34 PUBLICATIONS 52 CITATIONS

[SEE PROFILE](#)



Elisa Calabria

University of Verona

38 PUBLICATIONS 3,177 CITATIONS

[SEE PROFILE](#)



Enrico Tam

University of Verona

52 PUBLICATIONS 562 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Identification of functional biomarkers of frailty by measures of mitochondrial respiration and ROS production in blood cells. [View project](#)



I am writing the draft of a paper dealing with the effect of HIT and strength training on muscle architecture, neuromuscular activation and strength in active, elderly people. In addition, I am involved, as a supervisor, in a project of a PhD student focused on the peripheral and central determinants of maximal aerobic power in young adult and athletes [View project](#)

Maximal aerobic power and anaerobic capacity in cycling across the age spectrum in male master athletes

C. Capelli^{1,2} · J. Rittveger^{3,4} · P. Bruseghini¹ · E. Calabria¹ · E. Tam¹

Received: 24 November 2015 / Accepted: 16 May 2016
© Springer-Verlag Berlin Heidelberg 2016

Abstract

Purpose We analyzed the best performance times of master cycling athletes in the 200–3000 m track competitions to estimate the decay of maximal aerobic power (MAP) and anaerobic capacity (AnS) with aging.

Methods In various decades of age (30–80 years), MAP and AnS were estimated using an iterative procedure as the values that minimize the difference between: (1) the metabolic power ($\dot{E}(t)$) necessary to cover a given distance (d) in the time t and; (2) the maximal metabolic power ($\dot{E}_{\max}(t)$) maintained at a constant level throughout the competition.

Results MAP started decreasing at 45 years of age. Thereafter, it showed an average percent rate of decrease of about 16 % for decade, as previously shown in other classes of master athletes. In addition, AnS seemed to decay by about 11 % every 10 years from the second part of the fifth decade.

Conclusions The decay of MAP occurred in spite of the active lifestyle of the subjects and it may be attributed to the progressive impairment of maximal O_2 delivery and/or of peripheral O_2 utilization. The loss of AnS might derive

from the progressive loss of muscle mass occurring after the fifth decade of life, to the progressive qualitative deterioration of the anaerobic energy yielding pathways or to the lower capacity of MN recruitment during maximal efforts. The proposed approach may be applied to other types of human locomotion of whom the relationship between performance t and $\dot{E}(t)$ is known.

Keywords Master athletes · Maximal aerobic power · Anaerobic capacity · Cycling · Aging

Abbreviations

A	Frontal area of the subject riding the bike (m^2)
AnS	Anaerobic capacity (kJ)
[ATP]	Intramuscular adenosine-tri-phosphate concentration ($mM\ kg^{-1}$)
BPT	Best performance time (s)
C	Energy cost of human locomotion ($kJ\ km^{-1}, J\ m^{-1}\ kg^{-1}$)
$C_{c,a}$	Average energy cost of cycling during the acceleration phase for a stationary start ($kJ\ km^{-1}, J\ m^{-1}\ kg^{-1}$)
C_c	Energy cost of cycling, ($kJ\ km^{-1}, J\ m^{-1}\ kg^{-1}$)
C_{rr}	Rolling resistance coefficient
C_x	Drag coefficient
d	Distance (m)
$E_{acc,a}$	Amount of metabolic energy spent during the acceleration phase (kJ)
E_{Aer}	Percent contribution of aerobic energy sources to a given effort (%)
E_{AnS}	Percent contribution of anaerobic energy sources to a given effort (%)
$\dot{E}(t)$	Metabolic power required to cover a given distance d as a function of the time in human locomotion (kW)

Communicated by Jean-René Lacour.

✉ C. Capelli
carlo.capelli@univr.it; carlo.capelli@nih.no

¹ Department of Neurological and Movement Sciences, School of Exercise and Sport Sciences, University of Verona, Verona, Italy

² Department of Physical Performances, Norwegian School of Sport Sciences, Sognsveien 220, 0806 Oslo, Norway

³ German Aerospace Center (DLR), Institute of Aerospace Medicine, Cologne, Germany

⁴ Department of Pediatrics and Adolescent Medicine, University of Cologne, Cologne, Germany

$\dot{E}_c(t)$	Metabolic power required to cover a given distance d as a function of the time in cycling (kW)
$\dot{E}_{\max}(t)$	Maximal metabolic power available to the athlete as a function of the time of effort (kW)
η_c	Apparent mechanical efficiency of cycling
g	Acceleration of gravity (9.81 m s^{-2})
$\text{HR}_{\max}$	Maximal hear rate (beats per minute)
$[\text{La}]_b$	Blood lactate concentration, peak lactate concentration (mM)
M_t	Overall mass (subject plus frame) (kg)
MAP	Maximal aerobic power (kW)
$\text{MAP}_{1\text{H}}$	Maximal aerobic power calculated from the metabolic power maintained by during best hour unaccompanied performance (kW)
[PCr]	Intramuscular phosphocreatine concentration (mM kg^{-1})
ρ	Air density (kg m^{-3})
t_e	Time of exhaustion (s)
τ	Time constant of the mono-exponential increase of muscular oxygen uptake as a function of time of exercise (s)
s	Speed of locomotion (m s^{-1})
$\dot{V}\text{O}_{2\max}$	Maximal oxygen uptake (L min^{-1} , $\text{mL min}^{-1} \text{ kg}^{-1}$)

Introduction

Aging is the result of a continuous, irreversible process that evolves during the lifespan of most if not all organisms. From a biological point of view, the negative aspects of aging result from the accumulation of damage and structural modifications leading to the progressive impairment of several physiological functions. Among them, the decay of exercise capacity is of paramount importance, as it progressively limits the daily life autonomy and restricts physical activity in the aging subjects.

Maximal aerobic power (MAP)—related to maximal oxygen uptake ($\dot{V}\text{O}_{2\max}$)—is one of the most important physiological variables correlated with endurance and maximal performances and it is also a predictor of cardiovascular mortality (Lakka et al. 2002). After the peak in young adult age, MAP has been described to decay by about 10 % per decade in the general population after the age of 30 (Åstrand et al. 1973; Dill et al. 1967; Maharam et al. 1999; Marti and Howland 1990; Marcell et al. 2003; Robinson 1938; Rogers et al. 1990; Trappe et al. 1996), even though we do not know exactly at which age this decay becomes evident.

Exercise capacity, however, may also be limited, especially during very high intensity and supra-maximal efforts, by the amount of energy obtained from the anaerobic energy stores, i.e., the anaerobic capacity (AnS). In

this regard, the data describing the progressive drop of AnS with aging are rather scanty (Marti and Howland 1990). If we assume that it is mainly related to the muscle mass, however, we may surmise that it progressively decays in connection with a noticeable loss in the relative amount of muscle mass, which is progressively occurring after the fifth decade, i.e., 50 years of age (Janssen et al. 2000).

The rate of decay of MAP and AnS with age might be indirectly evaluated by analyzing the maximal speeds achieved by master athletes in different decades of age (Rittweger et al. 2009; Wiswell et al. 2001; Zamparo et al. 2012). For instance, the decline of endurance performance with age has been primarily ascribed to the progressive reduction of MAP and of the lactate threshold (Tanaka and Seals 2008). Conversely, as aging has been described to be paralleled by significant decays of anaerobic capacity and anaerobic power even in master athletes (Gent and Norton 2013), also best performances over shorter distances are likely bound to decrease with age. Other important factors affecting performances, such as exercise economy in the terrestrial forms of human locomotion, seems to be unaffected by aging (Tanaka and Seals 2008).

This analysis, however, could be further refined. In the recent past, several theoretical models to predict best performances in human locomotion running and cycling and swimming, e.g., have been proposed and tested (Capelli et al. 1998; di Prampero et al. 1993; Olds et al. 1993; Tous-saint and Hollander 1994). In general, these models are based on the fact that in human locomotion, the metabolic power $\dot{E}(t)$ required to cover a given distance d in the time t is set by the product of the energy cost of the locomotion at stake (C) and the speed ($v = d t^{-1}$), where C is the amount of metabolic energy spent to cover one unit of distance (di Prampero 1986). C of several forms of locomotion on land has been described in a quantitative fashion as a function of the terrain incline and type, the speed, the anthropometrical features of the subjects, the technical characteristics of the mean of locomotion, the environmental conditions, etc (Capelli et al. 1998; di Prampero et al. 1993; Minetti et al. 2002; Olds et al. 1993). Therefore, $\dot{E}(t)$ required to cover a given distance (d) as a function of t in the form of locomotion, such as running and cycling, can be individually determined, provided that we measure selected physiological and anthropometrical parameters of the subject and obtain the other essential parameters entering the equations.

Of course, the best performance time (BPT) over any given d satisfies also the equality $\dot{E}(t) = \dot{E}_{\max}(t)$, where the latter is the maximal metabolic power that is produced and maintained at a constant level by the athlete from the start of the competition to the end. $\dot{E}_{\max}$ is a decreasing function of t and mainly depends upon the subject's MAP and AnS (Capelli 1999; di Prampero et al. 1993).

Therefore, provided the relationship between C and s , and the subject's MAP and AnS are known, the best BPT over any given distance can be estimated by finding, through computerized iterative procedures, the value of t that solves the equality

$$\dot{E}_{\max}(t) = \dot{E}(t).$$

This approach, however, can also be reversed. In cycling, for instance, by knowing the BPT over several distances of a given athlete and by knowing his/her anthropometrical characteristics, the environmental conditions prevailing during the competition and the technical features of the frame, we may theoretically calculate backwards an athlete's MAP and AnS. Of course, this procedure, once applied to subjects of different ages, would allow us estimate the changes of MAP and AnS across the age spectrum without the interference of the additional confounding effects of disuse, detraining and sedentary life style, since master athletes keep training in view of participating to highly demanding competitions. Track cycling performances of Master Athlete over distances ranging from 200 to 3000 m seem to be particularly appropriate for this sort of analysis, because: (1) the shortest distances are covered performing high intensity exercises (<2 min), where anaerobic processes dominate; (2) the longest distance—3000 m is covered in less than 7 min, i.e., within an interval of time during which the cyclist is likely able to exploit throughout the effort a fraction of MAP equal to 1 (Péronnet and Thibault 1989). Therefore, by applying this line of reasoning, the backward resolution of the model may allow us to estimate the evolution of MAP and AnS across the age spectrum without the interference of the effects derived from assuming a sedentary life style or the presence of diseases. In addition, since it has been suggested that the rate of decay of running performances in the shorter distances may be significantly different from the one of the longer events (Baker et al. 2003; Fung and Ha 1994; Gent and Norton 2013; Moore 1975), this analysis may also unveil whether the decrease of MAP and AnS as a function of the age is similar or diverge.

In summary, the aim of this paper consists in estimating the changes of MAP and AnS in men master athletes participating in track cycling competitions across a spectrum of ages ranging from 40 to 80 years. MAP and AnS will be indirectly estimated (1) using BPT obtained by the master athletes over several distances competing in international events and classified in different classes of age and; (2) by finding the corresponding MAP and AnS values that solve the equality between $\dot{E}$ (BPT) and $\dot{E}_{\max}$ (BPT). This will also help understand whether the deterioration rates of these parameters with age are different.

Methods

Metabolic power requirement in track cycling

The metabolic power required covering as a function of time a given distance in track cycling has been calculated according to the approach proposed by Capelli (Capelli et al. 1998; Capelli 1999).

The total overall energy cost per unit of distance in track cycling (C_c) on flat terrain in the absence of wind is given by the sum of three terms:

$$C_c = C_{\pi} M_t g + k' s^2 + 0.5 M_t s^2 d_{\text{tot}}^{-1} \eta_c^{-1}, \quad (1)$$

where g is the acceleration due to gravity (m s^{-2}), s (m s^{-1}) is the air speed, which in still air can be considered identical to the ground speed, and the other terms are defined here below.

The first term ($C_{\pi} M_t g$) is the metabolic energy spent against rolling resistance; it is proportional to the product of the overall mass M_t (75 kg body mass, 10 kg frame) and to the coefficient C_{π} , which is the amount of energy spent over a unit of distance and per unit of the overall mass against frictional forces. The value of this term depends upon the characteristics of the tires and of the terrain and, for $M_t = 85$ kg and for $C_{\pi} = 0.031 \text{ J m}^{-1} \text{ kg}^{-1}$ (Capelli 1999), amounts to 25.8 J m^{-1} .

The second term ($k' v^2$) is the metabolic energy spent per unit of distance against air drag ($C_{c,a}$, J m^{-1}); it is proportional to the square of the air speed (s) and to a constant k' which, in turn, is described by (Capelli 1999):

$$k' = 0.5 C_x \rho A \eta_c^{-1}, \quad (2)$$

where C_x in Eq. 2 is the dimensionless drag coefficient, which can be considered as a constant at the range of speeds investigated (Pugh 1974); it was set equal to 0.58 for a cyclist riding a traditional racing bicycle in a fully dropped position (Capelli et al. 1993). The air density ρ for an air temperature of 20°C and a barometric pressure of 760 mmHg, disregarding the contribution of water vapor, amounts to 1.2047 kg m^{-3} ; A is the frontal area of the subject riding the bicycle, and amounts to $\approx 0.43 \text{ m}^2$ for a subject of 175 cm in stature and 75 kg body mass (Capelli et al. 1998). This yields the product of C_x times A of 0.249 m^2 (García-López et al. 2008). With an overall efficiency of cycling, η_c , equal to ≈ 0.22 (Seabury et al. 1977), k' turns out to be $0.68 \text{ J s}^2 \text{ m}^{-1} \text{ m}^{-2}$.

The speed becomes constant only after the acceleration phase has been completed; therefore, $C_{c,a}$ in the constant speed phase can be calculated as indicated by Olds and colleagues (1993) as:

$$C_{c,a} = 0.68 s^2 (d_{\text{tot}} - d_{\text{acc}}) d_{\text{tot}}^{-1} = 0.68 s^2 (1 - d_{\text{acc}}/d_{\text{tot}}), \quad (3)$$

where d_{tot} and d_{acc} represent: (1) the total distance and (2) the distance covered to accelerate from zero to the constant speed, assumed to be equal to 100 m (Olds et al. 1993).

Since the metabolic energy spent against drag depends on the instantaneous speed, $C_{c,a}$ during the acceleration phase is lower than that applied to the second part of the competition covered at constant speed. The overall value of $C_{c,a}$ may be corrected for the initial acceleration phase (100 m long) as follows. First, the total amount of metabolic energy spent against drag during the acceleration phase ($E_{\text{acc},a}$, J) may be calculated as:

$$E_{\text{acc},c} = 0.68s^2d_{\text{acc}}0.5, \quad (4)$$

where $E_{\text{acc},a}$ is the mean of the product of $C_{c,a}$ ($=0.68 \text{ s}^2$) times d_{acc} , calculated assuming that the acceleration was constant over d_{acc} (Olds et al. 1993). Then, the contribution of $E_{\text{acc},a}$ to the overall value of $C_{c,a}$ is calculated by dividing $E_{\text{acc},a}$ by the total distance d_{tot} . After some simplifications, the following equation is derived (Capelli 1999):

$$\begin{aligned} C_{c,a} &= 0.68s^2(1 - d_{\text{acc}}/d_{\text{tot}}) + 0.68s^2d_{\text{acc}}/d_{\text{tot}}0.5. \\ &= 0.68s^2(1 - 0.5d_{\text{acc}}/d_{\text{tot}}). \end{aligned} \quad (5)$$

The third term in Eq. 1 ($0.5 M_t s^2 d_{\text{tot}}^{-1} \eta_c^{-1}$) represents the metabolic energy spent by the cyclist to accelerate the overall mass M_t from a stationary start to the final speed s . Thus, if $M_t = 85 \text{ kg}$ and assuming $\eta_c = 0.22$ (Capelli et al. 1993; Seabury et al. 1977), the total metabolic requirement per-unit distance in track cycling (C_c , J m⁻¹) is described by:

$$C_c = 25.8 + 0.68s^2(1 - 0.5d_{\text{acc}}/d_{\text{tot}}) + 193.2s^2d_{\text{tot}}. \quad (6)$$

The overall metabolic power requirement necessary to progress at the speed v in cycling ($\dot{E}_c = C_c s$) is given by:

$$\dot{E}_c = 25.8s + 0.68s^3(1 - 0.5d_{\text{acc}}/d_{\text{tot}}) + 193.2s^3d_{\text{tot}}^{-1}. \quad (7)$$

Since in any track competition, the distance d_{tot} is fixed, and $d_{\text{tot}} t^{-1}$ can be substituted for s . As a consequence, Eq. 7 can be finally rearranged to obtain $\dot{E}_c$ as a function of the time necessary to cover the distance d_{tot} :

$$\begin{aligned} \dot{E}_c &= 25.8d_{\text{tot}}t^{-1} + 0.68d_{\text{tot}}^3t^{-3}(1 - 0.5d_{\text{acc}}/d_{\text{tot}}) \\ &+ 193.2d_{\text{tot}}^2t^{-3}. \end{aligned} \quad (8)$$

Maximal metabolic power

For competitions lasting less than 420 s (Péronnet and Thibault 1989), the maximal metabolic power ($\dot{E}_{\text{max}}$) a given subject can sustain at a steady level throughout the effort is a decreasing function of the exhaustion time (t_e) (Capelli et al. 1993; di Prampero et al. 1993) and it may be appropriately described as:

$$\dot{E}_{\text{max}} = \text{AnS}t_e^{-1} + [\text{MAP} - \text{MAP}\tau(1 - e^{-t_e\tau^{-1}})t_e^{-1}], \quad (9)$$

where AnS and MAP have been already defined, and τ is the time constant of the mono-exponential time course with which MAP is attained at the onset of the effort.

The third term arises from the fact that at the onset of exercise, $\dot{V}\text{O}_{2\text{max}}$ is not attained instantaneously, but with a time constant τ . Hence, the average aerobic power up to the time t_e is given by the quantity in square brackets, i.e., it is reduced below MAP by an amount equal to the oxygen deficit incurred up to t_e $\text{MAP}\tau(1 - e^{-t_e\tau^{-1}})$ divided by the time t_e itself. The first and third terms of Eq. 9 become progressively smaller with increasing values of t_e . Therefore, Eq. 9 shows that for long-term exercise, the maximal sustainable metabolic power is essentially set by the subjects $\dot{V}\text{O}_{2\text{max}}$ and by the fraction of $\dot{V}\text{O}_{2\text{max}}$ that can be maintained throughout the effort. As the time of the exercise becomes shorter, the contribution of the anaerobic energy stores to the overall metabolic power becomes progressively greater, because with decreasing t_e : (1) the first term of Eq. 9 increases hyperbolically and (2) the amount by which the oxygen deficit affects (negatively) the actual average aerobic power becomes larger.

Resolution of the model, data analysis, and statistics

For any given BPT obtained over a given distance, $\dot{E}_c$ (Eq. 8) required to cover d_{tot} as a function of BPT and $\dot{E}_{\text{max}}$ (BPT) (Eq. 9) must be equal: the record time corresponds to the unique condition wherein the required and the maximal available power are identical. Provided we have sufficient numbers of BPTs over different distances, we can calculate the values of MAP and AnS that minimize the differences between the couples of $\dot{E}_c$ and $\dot{E}_{\text{max}}$ on the various distances. In practice, an iterative procedure that implements the so-called Marquardt–Levenberg method for solving non-linear regression parameters (GraphPad Prism version 6.00 for Macintosh, GraphPad Software, La Jolla California USA, <http://www.graphpad.com>) (Levenberg 1944; Marquardt 1963) converges to the values of MAP and AnS that minimize the sum of squares of the differences between Eqs. 8 and 9 over several for various couples of the equality:

$$\begin{aligned} &25.8d_{\text{tot}}\text{BPT}^{-1} + 0.68d_{\text{tot}}^3\text{BPT}^{-3}(1 - d_{\text{acc}}/d_{\text{tot}} \\ &+ 0.5d_{\text{acc}}/d_{\text{tot}}) + 193.2d_{\text{tot}}^2\text{BPT}^{-3}, \\ &= \text{AnSBPT}^{-1} + [\text{MAP} - \text{MAP}\tau(1 - e^{-\text{BPT}\tau^{-1}})t_e^{-1}] \end{aligned} \quad (11)$$

where MAP and AnS are unknown.

In the present study, $\dot{E}_c$ over 200, 500, 750, 1000, and 3000 m in the 2012 and 2013 Men Master World Championships together with the world records over the same

distances were calculated for the following decades: 35–39, 40–44, 45–49, 50–54, 55–59, 60–64, 65–69, 70–74, and 75–80 year. Track cycling performances were obtained from the UCI Track Cycling masters World Championships web page (<http://www.cyclingmasters.com> accessed on May 2014). Best performances of the 200 m race were included, even though it is a flying start competition disregarding the fact that the athletes were not starting from a stationary start. For the calculations, the performances in the World Championships of the three first standings were always considered. In the youngest group of subjects, the performances over the 200 m in the 2013 World Championships were not considered for the calculations, since they were remarkably slower than the corresponding ones of the immediately older cyclists.

$\dot{E}_c$ was calculated by assuming for k' , C_{rr} , C_x , and η_c the values reported in the paragraphs above. M_t and A were assumed to be 85 kg and 0.43 m², respectively. ρ was calculated by assuming an air temperature of 20 °C and the average, standard barometric pressure (P_B) prevailing at the venue of the competition calculated using a standard formula that corrects P_B as a function of altitude above sea level (Haldane and Priestley 1935).

For each performance time, MAP–AnS values were calculated using a value of τ in Eq. 9 of 10 s, as proposed by di Prampero and colleagues (di Prampero et al. 1993). Finally, τ was not corrected for age, because in cycling master athletes, it was found not to increase with aging from the value found in the youngest subjects (Berger et al. 2006).

95 % confidence intervals of the estimated MAP and AnS were always calculated from the asymptotic standard errors (Motulsky and Christopoulos 2004). The goodness of fit was quantified using the coefficient of determination r^2 (Motulsky and Christopoulos 2004).

The threshold age after which MAP and AnS begin to decrease monotonically (age_0) has been identified fitting the absolute values of the two parameters as a function of the age (years) by means of a two-segment linear regression.

To this aim, a double linear regression that minimized the squared sum of the residuals was then fitted to the MAP and AnS data (GraphPad Prism version 6.00 for Macintosh, GraphPad Software, La Jolla California USA, <http://www.graphpad.com>):

$$f = \text{if } [x > \text{TD}, g(x), h(x)], \quad (12)$$

$$g(x) = i_1 + (s_1x) \quad (13)$$

$$i_2 = i_1 + (s_1\text{TD}) \quad (14)$$

$$h(x) = i_2 + s_2(x - \text{TD}) \quad (15)$$

fit f to y ,

where f is the double linear function, x is time, and y is either MAP or AnS; TD is the time coordinate corresponding to the interception of the two regression lines (the so-called threshold age); i_1 and i_2 are the intercepts of the first and second linear functions, respectively; and s_1 and s_2 are the slopes. i_2 was then considered to correspond to age_0 .

Results

Some master cycling events have variable distances, depending on age. For instance, single pursuit competition foresees a distance of 3000 m from 35 to 49 years athletes and of 2000 m for older athletes. Sprint distance, conversely, it is identical to every age, as it considers the time necessary to cover at the highest speed, the last 200 m from a flying start. Therefore, the analysis of the decay of v for a given event across the age spectrum considered in the present study (Table 1) is not possible as athletes of increasing decades inevitably compete over different distances.

On the contrary, it is possible to describe in a rather elegant way the decay of $\dot{E}_c$ (or $\dot{E}_{\text{max}}$) as a function of BPT in the nine considered intervals of age. In Fig. 1, for instance, the calculated values of $\dot{E}_c$ are plotted as a function of the corresponding BPTs for the intervals of age corresponding to the youngest, the intermediate age and the oldest athletes together with the values of MAP and AnS originated by the simulation. MAP and AnS appear, at least at first glance, to decrease with age. The same data are also reported in Table 2 with the addition of the 95 % ICs of the estimated parameters.

A more detailed representation of the decay of MAP and AnS as a function of age is shown in Fig. 2. MAP appears to remain stable until 40–44 years of age; thereafter, it is characterized by a steady rate of decay with age. The solution of the double segmental regression reveals that the MAP begins to decline at an $\text{age}_0 = 42.3 \text{ y} \pm 1.9$. Afterwards, it appears to decrease by $0.034 \text{ kW year}^{-1} \pm 0.002$ (0.34 kW per decade ± 0.02) with a drop of $\dot{V}\text{O}_{2\text{max}}$ per year of $0.10 \text{ L min}^{-1} \pm 0.01$ ($1.3 \text{ mL min}^{-1} \text{ kg}^{-1}$ per year ± 0.13 in our hypothetical subject of 75 kg of body mass). AnS seems to decrease after $\text{age}_0 = 49.7 \text{ y} \pm 5.3$ with an absolute average rate of decay equal to $0.94 \text{ kJ year}^{-1} \pm 0.14$ (9.4 kJ ± 1.40 per decade), corresponding to a drop of AnS expressed in equivalent of O_2 of $0.60 \text{ mL O}_2 \text{ kg}^{-1} \pm 0.09$ per year.

To compute the percent decay per year of MAP and AnS, we calculated the percent decreases of the two parameters setting to 100 % the average of the values prevailing in (1) the first two (35–39 and 40–44 year) and; (2) in the first three decades (35–39; 40–44 and 45–49) for MAP and AnS, respectively. The data were grouped considering that the decay of MAP and AnS became evident after about 45 and 48 years of age.

Table 1 Distance and speeds considered in the present for the calculation of MAP and AnS in the master cyclists together with the venue of the competition or record

Age (years)	Distance (m)	Speed (m s ⁻¹)	Event	Venue
35–39	200	18.17	World Championship 2012	Manchester (GBR)
		18.10		
		17.90		
		19.60		
	1000	15.07	World record	Colorado Spr. (USA)
		15.07		
		14.74	World Championship 2012	Manchester (GBR)
		14.68		
		15.81		
		14.78		
	3000	14.74	World Championship 2013	Manchester (GBR)
		15.40		
		14.03	World record	Colorado Spr. (USA)
		14.03		
		13.90	World Championship 2012	Manchester (GBR)
		13.89		
		14.09		
		14.06		
		14.02		
		14.93		
40–44	200	19.25	World record	Melbourne (AU)
		19.25		
		18.39	World Championship 2012	Manchester (GBR)
		17.75		
		16.16		
		15.62		
	750	15.59	World record	Manchester (GBR)
		19.29		
		15.11	World Championship 2012	Manchester (GBR)
		15.04		
		14.92	World Championship 2013	Manchester (GBR)
		15.33		
		15.11		
		14.82		
	3000	15.62	World record	Melbourne (AUS)
		14.63		
		14.57	World Championship 2012	Manchester (GBR)
		14.50		
		14.28	World Championship 2013	Manchester (GBR)
		14.27		
		13.87		
		14.64		
45–49	200	18.30	World record	Manchester (GBR)
		18.30		
		18.25	World Championship 2012	Manchester (GBR)
		17.78		
		18.08		
		17.96		
		17.84	World Championship 2013	Manchester (GBR)
		18.67		
		18.67	World record	Colorado Spr. (USA)

Table 1 continued

Age (years)	Distance (m)	Speed (m s ⁻¹)	Event	Venue
50–54	750	15.42	World Championship 2012	Manchester (GBR)
		14.98		
		14.68		
		15.34	World Championship 2013	Manchester (GBR)
		14.80		
		14.67		
	15.42	World record	Manchester (GBR)	
	3000	14.24	World Championship 2012	Manchester (GBR)
		14.23		
		14.14		
		14.02	World Championship 2013	Manchester (GBR)
		13.82		
		13.80		
	14.92	World record	Manchester (GBR)	
	200	17.87	World Championship 2012	Manchester (GBR)
		17.62		
		17.61		
		18.01	World Championship 2013	Manchester (GBR)
		17.75		
		17.56		
	18.03	World record	Manchester (GBR)	
	500	14.32	World Championship 2012	Manchester (GBR)
		14.25		
		14.15		
		14.53	World Championship 2013	Manchester (GBR)
		14.43		
		14.21		
2000	14.53	World record	Manchester (GBR)	
	14.25	World Championship 2012		
	14.24			
	13.98		World Championship 2013	
	14.27			
	14.19			
55–59	200	13.91	World record	Manchester (GBR)
		14.59		
		17.50		
		17.01	World Championship 2012	Manchester (GBR)
		17.01		
		17.50		
	500	17.34	World Championship 2013	Manchester (GBR)
		17.25		
		18.31		
		14.26	World Championship 2012	Colorado Spr. (USA)
		14.08		Manchester (GBR)
13.88				
500	14.35	World Championship 2013	Manchester (GBR)	
	14.18			
	14.12			
	14.56	World record	Manchester (GBR)	

Table 1 continued

Age (years)	Distance (m)	Speed (m s ⁻¹)	Event	Venue
60–64	2000	14.01	World Championship 2012	Manchester (GBR)
		13.68		
		13.42		
		13.80	World Championship 2013	Manchester (GBR)
		13.63		
		13.56		
		14.30	World record	Manchester (GBR)
	200	16.50	World Championship 2012	Manchester (GBR)
		16.28		
		16.18		
		16.64	World Championship 2013	Manchester (GBR)
		16.54		
		16.03		
		17.43	World record	Colorado Spr. (USA)
	500	13.50	World Championship 2012	Manchester (GBR)
		13.38		
		13.36		
		13.47	World Championship 2013	Manchester (GBR)
		13.43		
		13.36		
		14.02	World record	Colorado Spr. (USA)
	2000	13.46	World Championship 2012	Manchester (GBR)
		13.13		
		13.11		
		13.28	World Championship 2013	Manchester (GBR)
		13.25		
		12.85		
		13.81	World record	Colorado Spr. (USA)
65–69	200	16.23	World Championship 2012	Manchester (GBR)
		16.22		
		15.89		
		16.20	World Championship 2013	Manchester (GBR)
		16.01		
		15.51		
		16.87	World record	Colorado Spr. (USA)
	500	13.34	World Championship 2012	Manchester (GBR)
		13.00		
		12.83		
		13.35	World Championship 2013	Manchester (GBR)
		13.15		
		13.09		
		13.53	World record	Manchester (GBR)
	2000	12.91	World Championship 2012	Manchester (GBR)
		12.82		
		12.70		
		12.65	World Championship 2013	Manchester (GBR)
		12.63		
		12.58		
		13.48	World record	Manchester (GBR)

Table 1 continued

Age (years)	Distance (m)	Speed (m s ⁻¹)	Event	Venue
70–74	200	15.60	World Championship 2012	Manchester (GBR)
		15.09		
		15.05		
		15.89	World Championship 2013	Manchester (GBR)
		15.64		
		15.30		
	500	16.56	World record	Colorado Spr. (USA)
		12.96	World Championship 2012	Manchester (GBR)
		12.49		
		12.47		
		13.05	World Championship 2013	Manchester (GBR)
		12.83		
		12.15		
		13.39	World record	Manchester (GBR)
		12.39	World Championship 2012	Manchester (GBR)
		12.39		
		12.10		
	2000	12.19	World Championship 2013	Manchester (GBR)
		12.09		
		12.07		
		12.95	World record	Melbourne (AUS)
75–79	200	15.46	World record	Manchester (GBR)
		11.79	World Championship 2012	Manchester (GBR)
	500	11.14		
		10.88		
		12.10	World Championship 2013	Manchester (GBR)
		12.02		
		11.25		
		12.73	World record	Manchester (GBR)
	2000	11.57	World Championship 2012	Manchester (GBR)
		11.20		
		10.85		
		11.53	World Championship 2013	Manchester (GBR)
		11.39		
		10.87		
		11.70	World record	Manchester (GBR)

In Fig. 3, the percent decay of MAP is reported as a function of age together with the regression lines complemented with the 95 % confidence bands ($F = 234.0$; $P < 0.0001$; $r = 0.99$). The slopes of the regression lines indicate that MAP is characterized by a percent decay of $1.62 \% \pm 0.11$ from age₀ onwards. In the same Figure, the percent decreases of AnS are also shown together with the corresponding regression lines and their 95 % confidence bands ($F = 88.2$; $P < 0.0002$; $r = 0.97$). In this case, AnS would diminish with a rate of decay of $1.08 \pm 0.12 \%$ every year.

Figure 4 shows the percent contribution of aerobic (E_{Aer} %, upper panel) and anaerobic (E_{AnS} %, lower panel) energy sources as a function of the absolute best performance times in the decades of age considered in the study. E_{Aer} and E_{AnS} were calculated from the estimated values of MAP and AnS considering the corresponding performance time. The two diagrams indicate that the percent contributions of the two energy yielding pathways to the energy produced during the maximal effort obviously depended on the time of the trial; however, these were apparently unaffected by the age.

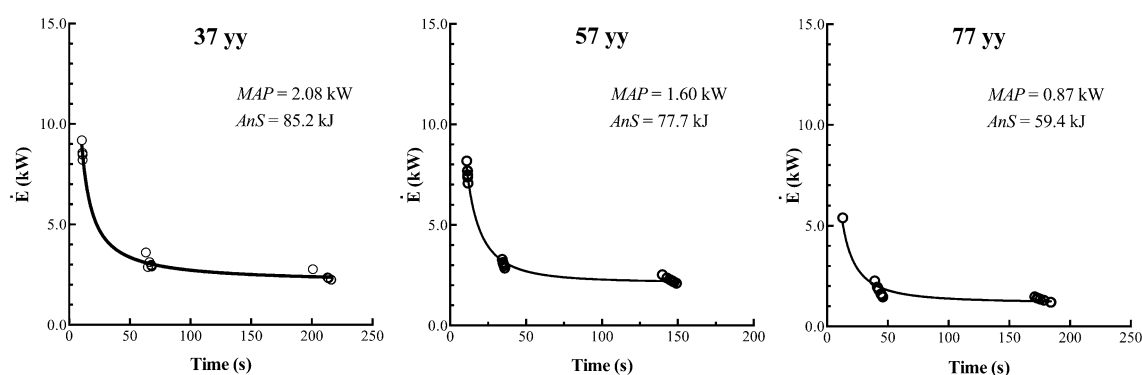


Fig. 1 $\dot{E}_c$ of master athletes as a function of performance times in the youngest, intermediate age, and oldest groups of subjects

Discussion

By applying a model utilized for predicting best performance in track cycling, we tried to describe the decay of MAP and AnS that occur with increasing age in cycling master athletes. This goal was achieved by estimating the values of MAP and AnS that minimized the difference between: (1) the metabolic power required by the subject to cover a given distance ($\dot{E}_c$) and (2) the maximal metabolic power output ($\dot{E}_{max}$) that he was able to maintain at a constant level throughout the effort. The applied method is similar to that applied by other investigators to estimate the decay with age of the maximal aerobic mechanical power and the amount of mechanical work performed using anaerobic energy sources in master, skyscraper runners (Minetti et al. 2011).

The results of the study suggest that MAP starts decreasing at 45 years of age in highly trained men. Thereafter, it shows an average percent rate of decrease of about 16 % for decade. In addition, AnS seems to decay by about 11 % per decade from the second part of the fifth decade.

Maximal aerobic power and age

Several papers have demonstrated that MAP decreases with aging in sedentary and active men as well as in well-trained male master athletes (Åstrand et al. 1973; Dill et al. 1967; Maharam et al. 1999; Marcell et al. 2003; Marti and Howland 1990; Pollock et al. 1997; Robinson 1938; Trappe et al. 1996). It has also been suggested that MAP decreases, on the average, by about 10 % per decade in the general population, even though values ranging from 5 to 15 % of variation every 10 years have been found for men of 20–75 years (Maharam et al. 1999). Conversely, the rate of decrease for master athletes has been suggested to be half of that of their sedentary mates (Kasch et al. 1995; Proctor and Joyner 1997; Rogers et al. 1990; Trappe et al. 1996).

However, whether the rate of decay of MAP with age in Master Athletes is lower or identical to the one found in active or untrained mates remains to be elucidated. For instance, when only older fit subjects were evaluated at 68 year of age, the rate of decrease of MAP was equal to 15 % per decade, i.e., identical to that of the untrained men (Trappe et al. 1996). In addition, Polloch and colleagues (Pollock et al. 1997), during their 20-year period of longitudinal observation, found a loss of only 8 % the first 10 years (from 50.5 to 60.2 years, on the average) followed to an additional loss of 15 % in the second decade (from 60.2 to 70.4, years, on the average), i.e., the rate of decline seemed almost double at the older age despite continued vigorous endurance during the older decades.

Therefore, in spite of contrasting results, the rate of reduction of MAP seems to accelerate in the advanced decades of age and this higher rate of decrease cannot be delayed or contrasted by training.

Our data seem to confirm this view, since MAP seemed to be preserved up to about 45 years of age in our master athletes. This is somehow in agreement with the small rate of decay found by other investigators in the same span of age. After this age, however, it decreased according to a percent rate of decline close to the one calculated by Wiswell et al. (1.2 percent decay per year) in a group of 146 master athletes (Wiswell et al. 2001) and by Johnson and colleagues (about 1.5 % decay per year) (Johnson et al. 2000). This seems to confirm that, after about the age of 50, the MAP progressively decreases despite continuous aerobic training.

Several physiological changes occurring with aging may be advocated as causes of the progressive drop of MAP. Theoretically, the progressive impairments of both maximal bulk oxygen delivery and of peripheral O_2 diffusion-utilization may contribute to the observed decay of MAP. HR_{max} has been described to decrease by about 1 beat per year after 10 years of age. Although a decline of HR_{max} has also been shown in master, well-trained athletes, its rate of

Table 2 Maximal aerobic power (MAP) and anaerobic capacity (AnS) of master athletes in the 95-year interval of ages considered in the study

Age (years)	MAP (kW)	AnS (kJ)
35–39	2.08	85.2
95 % CI	(1.94–2.12)	(82.8–87.7)
40–44	2.12	80.7
95 % CI	(1.54–2.69)	(72.3–89.1)
45–49	1.99	83.9
95 % CI	(1.816–2.17)	(81.36–86.5)
45–49	1.76	81.3
95 % CI	(1.52–2.00)	(77.9–84.7)
55–59	1.60	77.7
95 % CI	(1.39–1.84)	(74.7–80.7)
60–64	1.40	67.2
95 % CI	(1.18–1.62)	(64.0–70.4)
65–69	1.33	64.0
95 % CI	(1.13–1.54)	(61.0–67.0)
70–74	1.18	60.2
95 % CI	(0.97–1.39)	(56.9–63.4)
75–79	0.87	59.4
95 % CI, $\tau_{25\text{ s}}$	(0.61–1.12)	(51.2–67.6)

For more details, the reader is kindly asked to refer to the text

decay seems to be slightly lower (a loss of about 4–7 beats per decade) (Kasch et al. 1995; Maharam et al. 1999; Rogers et al. 1990; Pollock et al. 1997; Trappe et al. 1996) than the one observed in the general population. The progressive drop of HR_{max} would imply a parallel decay of maximal oxygen delivery that would well explain the concurrent drop of MAP. Less is known about the changes of the ventricular performances with age. According to (Douglas and O'Toole 1992), several cardiac functions, including greater posterior wall thickness, lower rapid filling velocity, higher atrial systolic velocity, and lower early-to-atrial inflow velocity ratios, differ in master athletes in respect to younger subjects. Master athletes are also characterized by decreased left ventricular compliance compared with the young control subjects, in the presence of higher cardiac pressures for a given filling volume and of higher myocardial wall stress for a given strain (Arbab-Zadeh et al. 2004). However, the increase of stroke volume during exercise in response to identical increase in end-diastolic volume is significantly larger in master athletes than in controls. This indicates that enhanced left ventricular systolic function independent of preload preserves a higher stroke volume at peak exercise and permits to attain a larger stroke volume (Seals et al. 1994). The decay of MAP with age may also be the consequence of the progressive impairment of peripheral gas exchanges and of muscular oxidative

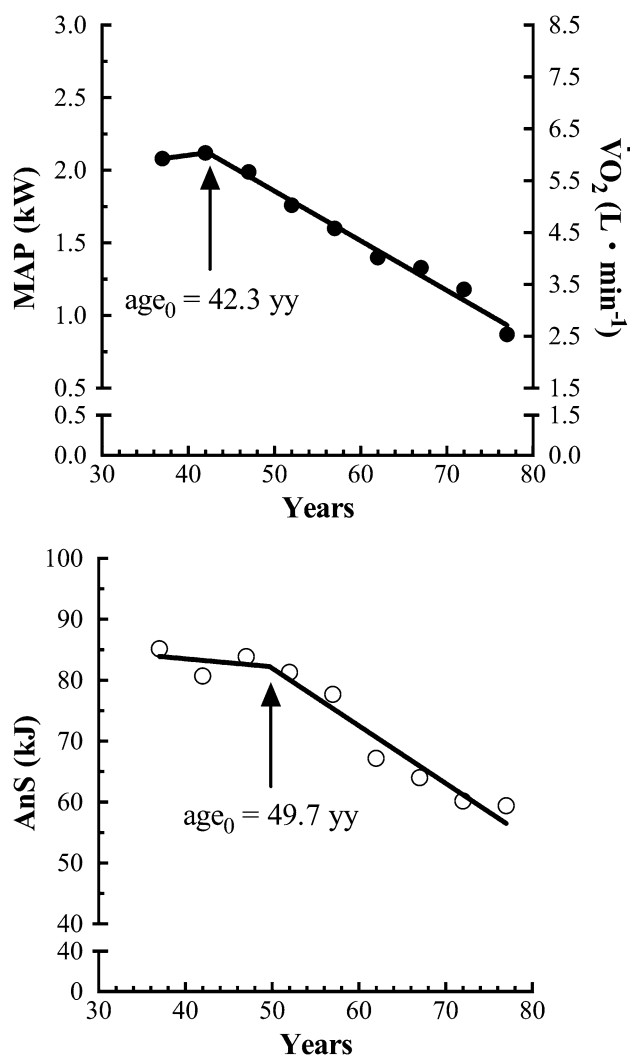


Fig. 2 Absolute values of MAP (*upper panel*) and AnS (*lower panel*) as a function of age. *Arrows* indicate the age (age_0) after which MAP and AnS start monotonically decreasing. For MAP, the right vertical axis reports also the scale in $\text{L O}_2 \text{ min}^{-1}$

metabolism. $\dot{V}\text{O}_{2\text{max}}$ normalized per kg of appendicular muscle was found to decay in trained old subjects (Proctor and Joyner 1997), so that the decline of $\dot{V}\text{O}_{2\text{max}}$ with age may be only partially explained by the concomitant loss of skeletal muscle mass. However, there seems not to be conclusive evidence on the role of the progressive loss of muscular oxidative metabolic capacities on MAP. Although we know that oxidative enzyme activity (Proctor et al. 1995), capillarization (Coggan et al. 1992), mitochondrial density, and oxidative capacity decay with age (Conley et al. 2000; Lanza and Nair 2010), several studies have found similar muscular oxidative capacity in trained young and older adults (Lanza and Nair 2010) and increased mitochondrial biogenesis and ETC activity in trained older subjects

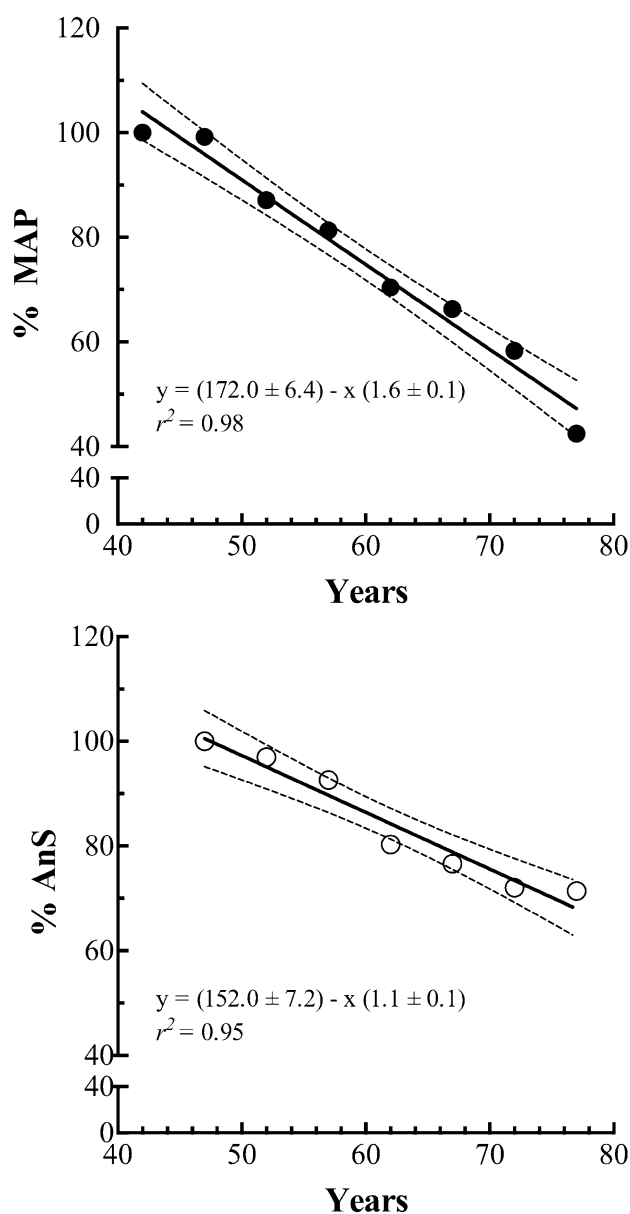


Fig. 3 Percent decrease of MAP (*upper panel*) and AnS (*lower panel*) as a function of age after age₀. The regression lines with their 95 % confidence bands are also indicated

(Menshikova et al. 2006). Finally, also the loss of fat-free body mass (FFM) may substantially contribute to the age related loss of MAP, as regression models showed that FFM decline explains more the 60 % of the variation of MAP with age (Johnson et al. 2000).

Anaerobic capacity and age

AnS was shown to significantly decrease with age after the second half of the fifth decade in the present study (Fig. 3).

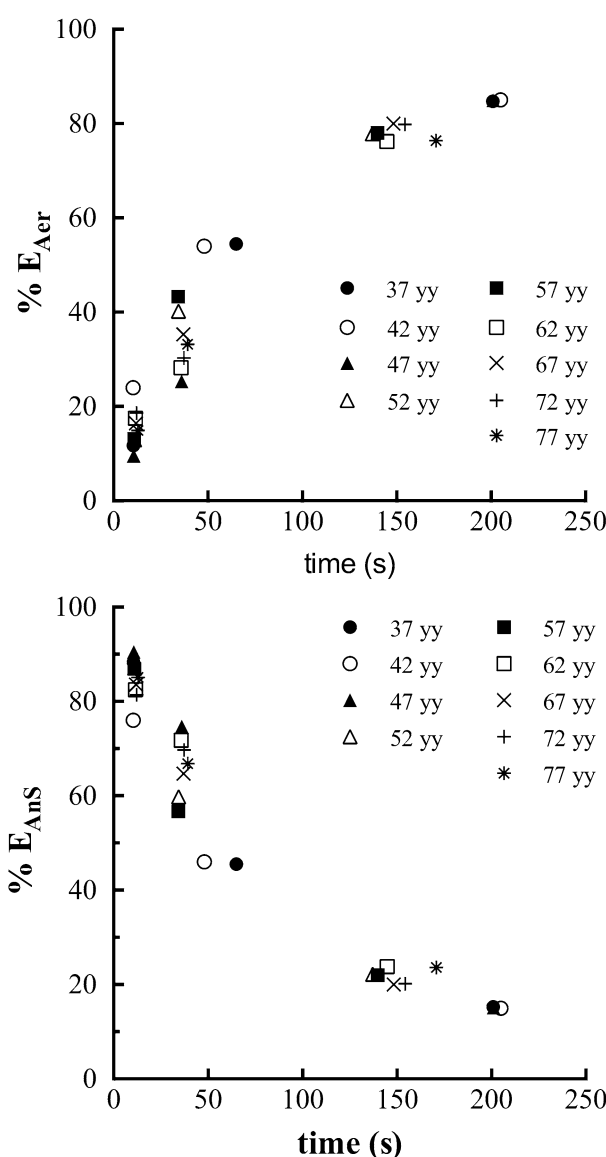


Fig. 4 Percent contribution of aerobic (E_{Aer} %, *upper panel*) and anaerobic (E_{AnS} %, *lower panel*) energy sources as a function of the absolute best performance time in the different groups of age

Korhonen and colleagues (Korhonen et al. 2005) measured peak blood lactate concentration ($[La]_b$) in master male runners after sprint events and showed that it declined in a curvilinear fashion with age. Marsh (Marsh et al. 1999) showed that peak and mean power during a lower-limb Wingate test, as well as $[La]_b$, were significantly lower in active 70 years old subjects than in active, younger mates. During high-intensity, supra-maximal efforts, also the anaerobic alactic fraction of total anaerobic capacity, namely, the amounts of muscular PCr and ATP available to breakdown supplies a substantial amount of energy. In the present study, the estimated value of AnS would obviously

include also the alactic counterpart. It has been shown that intramuscular [ATP] progressively decays with age in healthy, non-athletic subjects, whereas intramuscular [PCr] remains unchanged (Kersick et al. 2016), but no data of this kind seem to be available for master athletes.

The aforementioned drop of AnS observed with age may be connected both to quantitative and qualitative muscular factors.

The loss of muscle mass with age has been broadly documented (Lexell 1993; Narici and Maffulli 2010) and it is greater in the lower than in the upper limbs (Janssen et al. 2000). Although skeletal muscle mass starts declining in the third decade, a remarkable decline of muscle mass is not detected until the second part of the fifth decade of age in healthy elderly (Janssen et al. 2000) and a significant decline of *FFM* with age have also been observed in master running athletes (Pollock et al. 1997; Trappe et al. 1996). The reduction of skeletal muscle mass volume and cross-sectional area are explained by the decline of fiber size (atrophy) and number (sarcopenia) (Narici and Maffulli 2010). The bulk of evidence shows that Type II muscle fibers are more liable to atrophy with age (Narici and Maffulli 2010) also in active, sprint male master athletes (Korhonen et al. 2006).

The drop in AnS and the performance in short-term supra-maximal events observed with age could also be related to qualitative muscular changes. For instance, a shift toward slower myosin isoform profiles even in sprint master athletes (Korhonen et al. 2006). This may obviously contribute to the decay of performance over short distance, supra-maximal events. From the biochemical standpoint, increasing age is characterized by a decrease of phosphofructokinase-1 the allosterically controlled enzyme of the key reaction of glycolysis (Evans et al. 1992); this may substantially lower the flux of substrate through the glycolytic pathway and the substrate-level phosphorylation rate of ATP synthesis.

In addition, the neuromuscular functions may undergo substantial detrimental changes with aging associated with a reduction of the number and diameter of large-diameter motoneuron axons in the ventral leading to a drop of the speed of conduction axonal conduction speed (Aagard et al. 2010) that becomes already evident after the fifth-sixth decade of age (Mittal and Logmani 1987). It is worth noting that Grassi and colleagues attributed the deterioration of maximal muscular power of the lower found up to the age of 45 years to qualitative muscular features, whereas the subsequent decay was mainly explained by the loss of muscle mass (Grassi et al. 1991).

In conclusion, several muscular modifications occurring with age seem to justify and explain the observed decay of AnS and explain the decay of power during supra-maximal, short distance events. Finally, they also seem in agreement

with the suggestion that AnS appears to monotonically decay after approximately the end of the fifth decade, at least in this category of master athletes.

Points of weakness and strength

The model used in the present study is based on the equivalence between the metabolic power necessary to cover a given distance as a function of time and the corresponding total maximal power that the subject is able to provide in the same interval of time. This sort of models has been successfully utilized in the past to predict best performance times in running (Péronnet and Thibault 1989; di Prampero et al. 1993) and in track cycling (Capelli et al. 1998; Olds et al. 1993) and to evaluate the impact of environmental conditions (Minetti et al. 2002; Péronnet et al. 1991) and physiological factors (Capelli 1999) on maximal human performances. In all the cases, this approach has proved to estimate theoretical best performance times with remarkable precision both in elite and medium level, young athletes (Capelli et al. 1998; Olds et al. 1993; Péronnet and Thibault 1989; di Prampero et al. 1993). Therefore, it seems to be based on a sound knowledge of the physiological mechanisms dictating metabolic energy production in humans and of the bioenergetical aspects of human locomotion.

The model at stake assumed that the total amount of energy coming from anaerobic capacity was fully available independently of the time of exercise. This assumption has been questioned by some authors who proposed that it increases with the duration of exhausting exercise with a time constant of 23.4 s to attain a constant value only for exercises lasting longer than 120 s (Medbø and Tabata 1993). Therefore, the estimated values of AnS may somehow overestimate the real ones, as no correction for the time of exercise has been introduced. Conversely, the values of MAP may be considered to be representative of $\dot{V}O_{2max}$ as the time duration of the longer track events was shorter than 7 min, the time limit is still valid for considering F equal to 1. The model also assumes a fixed value of the time constant τ of the mono-exponential increase of muscular O_2 uptake at the onset of maximal exercise equal to 10 s, as proposed in the past for similar aims (di Prampero et al. 1993). τ of the Phase II of alveolar O_2 uptake is considered as a reliable proxy of the muscular $\dot{V}O_2$ kinetics and it amounts 22–25 s in healthy, trained humans during moderate intensity exercise (Poole et al. 2008). Therefore, the assumed value of 10 s may substantially overestimate the speed of adjustment of the aerobic metabolism even in well-trained athletes. We have, however, to consider that in the present case, we deal with supra-maximal or maximal exercise conditions. Shorter τ of the Phase II were found in supra-maximal than during moderate intensity exercise (Adami et al. 2012). A tentative explanation of this

phenomenon may be as follows. Although $\dot{V}O_2$ seems to project toward the metabolic requirement imposed by the external workload, this value cannot be attained, since it exceeds $\dot{V}O_{2\max}$. Hence, $\dot{V}O_2$ should increase following a kinetic identical to that prevailing during moderate exercise, an “apparently” shorter τ would inevitably result (Adami et al. 2012). It is worth noting that Minetti and colleagues (Minetti et al. 2011), who estimated the decay of MAP, ANS, and τ in master skyscraper runners by applying an approach similar to the one proposed in the present investigation, found values of τ ranging from 5 to 9 s in athletes 45–65 years old. However, the small value of τ adopted in this occasion may have led to the underestimation of the values of AnS (Table 2), which equaled to 55 mL O₂ kg⁻¹ in the hypothetical youngest master athletes. Yet, the AnS value estimated for these athletes were of the same order of magnitude of the ones reported for younger trained cyclists (56 to 81 mL O₂ kg⁻¹) (Gastin 1994).

We also know that τ increases with age in moderately active subjects (DeLorey et al. 2004). Conversely, it did not significantly changed with aging in endurance trained master athlete (Berger et al. 2006). Therefore, the assumption of an invariant τ seems to be tenable in applying our model.

The calculation of the metabolic power required covering a given d also depends on the total mass (Eq. 1), which obviously included the subject's body mass. Indeed, muscle mass and lean body mass have been found to decrease with age even in master athletes (Fleg and Lakatta 1988; Wiswell et al. 2001). However, total body mass in master athletes seems to be preserved across a broad spectrum of age (Trappe et al. 1996; Wiswell et al. 2001).

Of course, as the calculations are crucially based on best performance times, the reliability of the database of these values is the uppermost requirement for this sort analysis. These values might be not fully illustrative of the physiological features of some categories of master athletes where world championship competitions are not yearly organized and where absolute best performances may not be systematically recorded.

The main point of strength of study consists in the attempt of estimating with an indirect approach, not only the decay of MAP across the age spectrum—which has been directly assessed in several cross-sectional or longitudinal investigations in the past—but also that of anaerobic capacity, i.e., of an additional physiological parameter determining human performances during short-term, supra-maximal efforts. This approach may be applied to all the types of human locomotion—running (Rittweger et al. 2009), swimming (Zamparo et al. 2012), kayaking, canoeing, cross country skiing, etc—of whom the relationship between the speed and the corresponding C is known and described in terms of terrain inclination, type of surface,

technical tools utilized for moving or applied techniques, etc. As such, it would allow us obtain an overall and broader picture of the decay of human exercise capacity, and of its energetic determinants, with aging.

Conclusions

In conclusion, by applying a theoretical model for predicating human best performances in track cycling, we estimated the decay of MAP and AnS of male master athletes. These two physiological parameters are important factors determining human performances (Capelli 1999) and are affected in a multifactorial fashion by aging. The analysis suggests that they initiate to drop from the fifth decade of age. Afterwards, MAP declines by about 16 % per decade, as previously shown in other classes of master athletes. In addition, AnS shows a linear decay of about 11 % per decade, which might be mainly related to the progressive loss of muscle mass in the lower limbs and to the progressive deterioration of the anaerobic energy yielding pathways and neuromuscular characteristics.

Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest. The study was funded by the FUR 2014—UNIVR allocated to Carlo Capelli and Enrico Tam by the University of Verona.

References

- Aagard P, Suetta C, Caserotti P, Magnusson SP, Kjær M (2010) Role of the nervous system in sarcopenia and muscle atrophy with aging: strength training as a countermeasure. *Scand J Med Sci Sports* 20:49–64
- Adami A, Pogliaghi S, De Roia G, Capelli C (2012) Oxygen uptake, cardiac output and muscle deoxygenation at the onset of moderate and supramaximal exercise in humans. *Eur J Appl Physiol* 111:1517–1527
- Arbab-Zadeh A, Dijk E, Prasad A, Fu Q, Torres P, Zhang R, Thomas JD, Palmer D, Levine BD (2004) Effect of aging and physical activity on left ventricular compliance. *Circulation* 110:1799–1805
- Åstrand I, Åstrand PO, Hallback I, Kilbom Å (1973) Reduction in maximal oxygen uptake with age. *J Appl Physiol* 35:649–654
- Baker BA, Tang YQ, Turner MJ (2003) Percentage decline in master superathlete track and field performance with aging. *Exp Aging Res* 29:47–65
- Berger NJA, Rittweger J, Kwiet A, Michaelis I, Williams AG, Tolfrey K, Jones AM (2006) Pulmonary O₂ uptake kinetics in endurance and sprint-trained master athletes. *Int J Sports Med* 27:1005–1012
- Capelli C (1999) Physiological determinants of best performances in human locomotion. *Eur J Appl Physiol* 80:298–307
- Capelli C, Rosa G, Butti F, Ferretti G, Veicsteinas A, di Prampero PE (1993) Energy cost and efficiency of riding “aerodynamic bicycles”. *Eur J Appl Physiol* 67:144–149

- Capelli C, Schena F, Zamparo P, Dal Monte A, Faina M, di Prampero PE (1998) Energetics of best performances in track cycling. *Med Sci Sports Exerc* 30:614–624
- Coggan AR, Spina RJ, King DS, Rogers MA, Brown M, Nemeth PM, Holloszy JO (1992) Skeletal muscle adaptations to endurance training in 60–70-year old men and women. *J Appl Physiol* 72:1780–1786
- Conley KE, Jubrias SA, Esselman PC (2000) Oxidative capacity in human muscle. *J Physiol* 526:203–210
- DeLorey DS, Kowalchuk JM, Paterson DH (2004) Effect of age on O_2 uptake kinetics and the adaptation of muscle deoxygenation at the onset of moderate-intensity cycling exercise. *J Appl Physiol* 97:165–172
- di Prampero PE (1986) The energy cost of human locomotion on land and in water. *Int J Sports Med* 7:55–72
- di Prampero PE, Capelli C, Pagliaro P, Antonutto G, Girardis M, Zamparo P, Soule RG (1993) Energetics of best performances in middle distance running. *J Appl Physiol* 74:2318–2342
- Dill DB, Robinson S, Ross JC (1967) A longitudinal study of 16 champion runners. *J Sports Med Phys Fitness* 7:4–27
- Douglas P, O'Toole M (1992) Aging and physical activity determine cardiac structure and function in the older athlete. *J Appl Physiol* 72:1969–1973
- Evans L, Rice CL, Noble EG, Paterson DH, Cunningham DA, Taylor AW (1992) Comparison of histochemical, biochemical and contractile properties of triceps surae of trained aged subjects. *Can J Aging* 11:412–425
- Fleg JL, Lakatta EG (1988) Role of muscle loss in the age-associated reduction of $\dot{V}O_{2\max}$. *J Appl Physiol* 65:1147–1151
- Fung L, Ha A (1994) Changes in track and field performance with chronological aging. *Int J Aging Hum Dev* 38:171–180
- García-López J, Rodríguez-Marroyo JA, Juneau CE, Peleteiro J, Córdova Martínez A, Villa JG (2008) Reference values and improvement of aerodynamic drag in professional cyclists. *J Sports Sci* 26:277–286
- Gastin PB (1994) Quantification of anaerobic capacity. *Scand J Med Sci Sports* 4:91–112
- Gent DB, Norton K (2013) Aging has greater impact on anaerobic versus aerobic power in trained master athletes. *J Sports Sci* 31:97–103
- Grassi B, Cerretelli P, Narici MV, Marconi (1991) Peak anaerobic power in master athletes. *Eur J Appl Physiol* 62:394–399
- Haldane JS, Priestley JG (1935) *Respiration*, 2nd edn. Yale University Press, New Haven, p 323
- Janssen I, Heymsfield SB, Wang Z, Ross R (2000) Skeletal muscle mass and distribution in 468 men and women aged 18–88 year. *J Appl Physiol* 89:81–88
- Johnson PJ, Winter EM, Paterson DH, Koval JJ, Nevill AM, Cunningham DA (2000) Modeling the influence of age, body size and sex on maximum oxygen uptake in older humans. *Exp Physiol* 85:219–225
- Kasch FW, Boyer JL, Van Camp S, Nettl F, Verity LS, Wallace JP (1995) Cardiovascular changes with age and exercise. A 28-year longitudinal study. *Scand J Med Sci Sports* 5:147–151
- Kersick CM, Roberts MD, Dalbo VJ (2016) Intramuscular phosphagen status and the relationship to muscle performance across the age spectrum. *Eur J Appl Physiol*. 1:115–127. doi:10.1007/s00421-015-3246-1
- Korhonen MT, Suominen H, Mero A (2005) Age and sex differences in blood lactate response to sprint running in elite master athletes. *Can J Appl Physiol* 30:647–665
- Korhonen MT, Cristea A, Alén M, Häkkinen K, Sipilä S, Mero A, Viitasalo JT, Larsson L, Suominen H (2006) Aging, muscle fiber type, and contractile function in sprint-trained athletes. *J Appl Physiol* 101:906–917
- Lakka HM, Laaksonen DE, Lakka TA, Niskanen LK, Kumpusalo E, Tuomilehto J, Salonen JT (2002) The metabolic syndrome and total and cardiovascular disease mortality in middle-aged men. *J Am Med Assoc* 288:2709–2716
- Lanza IR, Nair KS (2010) Mitochondrial function as a determinant of life span. *Pflugers Arch Eur J Physiol* 459:277–289
- Levenberg K (1944) A method for the solution of certain non-linear problems in least squares. *Q Appl Math* 2:164–168
- Lexell J (1993) Ageing and human muscle: observations from Sweden. *Can J Appl Physiol* 18:2–18
- Maharam LG, Bauman PA, Kalman D, Skolnik H, Perle SM (1999) Master athletes. Factors affecting performances. *Sports Med* 28:273–285
- Marcell TJ, Hawkins SA, Trapenning KM, Hyslop DM, Wiswell RA (2003) Longitudinal analysis of lactate threshold in male and female master Athletes. *Med Sci Sports Exerc* 35:810–817
- Marquardt DW (1963) An algorithm for least-squares estimation of nonlinear parameters. *J Soc Ind Appl Math* 11:431–444
- Marsh GD, Paterson DH, Govindasamy D, Cunningham DA (1999) Anaerobic power of the arms and legs of young and older men. *Exp Physiol* 84:589–597
- Marti B, Howland H (1990) Long-term effects of physical training on aerobic capacity: controlled study of former elite athletes. *J Appl Physiol* 69:1451–1459
- Medbø JL, Tabata I (1993) Anaerobic energy release in working muscle during 30s to 3 min of exhausting exercise. *J Appl Physiol* 75:1654–1660
- Menshikova EV, Ritov VB, Fairfull L, Ferrell RE, Kelley DE, Goodpastre BH (2006) Effects of exercise on mitochondrial content and function in aging human skeletal muscle. *J Gerontol* 61:534–540
- Minetti AE, Moia C, Roi GS, Susta D, Ferretti G (2002) Energy cost of walking and running at extreme uphill and downhill slopes. *J Appl Physiol* 93:1039–1046
- Minetti AE, Cazzola D, Seminati E, Giacometti M, Roi GS (2011) Skyscraper running: physiological and biomechanical profile of a novel sport activity. *Scand J Med Sci Sports* 21:293–301
- Mittal KR, Logmani FH (1987) Age-related reduction in 8th cervical ventral nerve root myelinated fiber diameters and numbers in man. *J Gerontol* 42:8–10
- Moore DH (1975) A study of age group track and field records to relate age and running speed. *Nature* 253:264–265
- Motulsky HY, Christopoulos A (2004) *Fitting models to biological data using linear and nonlinear regression*. Oxford University Press, Oxford
- Narici MV, Maffulli V (2010) Sarcopenia: characteristics, mechanisms and functional significance. *Br Med Bull* 95:139–159
- Olds TS, Norton KI, Craig NP (1993) Mathematical model of cycling performance. *J Appl Physiol* 75:730–737
- Péronnet F, Thibault G (1989) Mathematical analysis of running performance and world running records. *J Appl Physiol* 67:453–465
- Péronnet F, Thibault G, Cousineau DL (1991) The theoretical analysis of the effect of altitude on running performance. *J Appl Physiol* 70:399–404
- Pollock ML, Mengelkoch LJ, Graves JE, Lowenthal DT, Limacher MC, Foster C, Wilmore JH (1997) Twenty-year follow-up of aerobic power and body composition of older track athletes. *J Appl Physiol* 82:1508–1516
- Poole DC, Barstow TJ, Mc Donough P, Jones AM (2008) Control of oxygen uptake during exercise. *Med Sci Sports Exerc* 40:462–474
- Proctor DN, Joyner MJ (1997) Skeletal muscle mass and the reduction of $\dot{V}O_{2\max}$ in trained older subjects. *J Appl Physiol* 82:1411–1415
- Proctor DN, Sinning WE, Walro JM, Sieck GC, Lemon PWR (1995) Oxidative capacity of human muscle fiber types: effects of age and training status. *J Appl Physiol* 78:2033–2038
- Pugh LGCE (1974) The relation of oxygen intake and speed in competition cycling and comparative observations on the bi-cycle ergometer. *J Physiol (Lond)* 241:795–808

- Rittweger J, di Prampero PE, Maffulli N, Narici MV (2009) Sprint and endurance power and ageing: an analysis of master athletic world record. *Proc R Soc B* 276:683–689
- Robinson S (1938) Experimental studies of physical fitness in relation to age. *Arbeits Physiol* 10:251–323
- Rogers MA, Hagberg JM, Martin WH III, Ehsani AA, Holloszy JO (1990) Decline in $\dot{V}O_{2\max}$ with aging in master athletes and sedentary men. *J Appl Physiol* 68:2195–2199
- Seabury JJ, Adams WC, Ramey MR (1977) Influence of pedaling rate and power output on energy expenditure during bicycle ergometry. *Ergonomics* 20:491–498
- Seals DR, Hagberg JM, Spina RJ, Rogers MA, Schechtman KB, Ehsani AA (1994) Enhanced left ventricular performance in endurance trained older men. *Circulation* 89:198–205
- Tanaka H, Seals DR (2008) Endurance exercise performance in Master athletes: age-associated changes and underlying physiological mechanisms. *J Physiol* 586:55–63
- Toussaint HM, Hollander AP (1994) Energetics of competitive swimming. Implications for training programmes. *Sports Med* 18:384–405
- Trappe S, Costill D, Vukovich M, Jones J, Melham T (1996) Aging among elite distance runners: a 22-year longitudinal study. *J Appl Physiol* 80:285–290
- Wiswell RA, Hawkins SA, Jaque SV, Hyslop DH, Constantino N, Trappenning K, Marcell T, Schroeder ET (2001) Relationship between physiological loss, performance decrement, and age in master athletes. *J Gerontol* 56:M618–M626
- Zamparo P, Gatta G, di Prampero PE (2012) The determinants of performance in master swimmers: an analysis of master world records. *Eur J Appl Physiol* 112:3511–3518